Selective State Space Models: A Linear-Time Alternative to Attention

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Abstract

Selective state space models (SSMs), popularized by Mamba, process sequences in linear time by replacing quadratic self-attention with an input-dependent recurrence. This submission teaches the core mechanism behind selective SSMs, discretized linear recurrences, the selection mechanism, and the hardware-aware parallel scan that makes them fast, to learners with a first course in linear algebra and deep learning. The accompanying notebook builds a minimal selective-scan layer from scratch, contrasts it against a naive recurrent implementation and a self-attention block, and visualizes how the model's hidden state evolves and selectively forgets across a sequence.

1 The concept

Transformers scale quadratically with sequence length because self-attention compares every token to every other token. State space models (SSMs) instead maintain a fixed-size hidden state $h_t \in \mathbb{R}^N$ that summarizes the sequence so far, updated by a linear recurrence

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \qquad y_t = Ch_t, \tag{1}$$

where $\bar{A}, \bar{B}$ are discretized versions of continuous-time matrices A, B (via a step size Δ), inherited from classical control theory and the structured/HiPPO-initialized SSMs (S4) that preceded Mamba.

The key limitation of earlier SSMs was that A, B, C, Δ were fixed for the whole sequence, so the model could not selectively attend to or ignore specific tokens. Mamba's contribution is to make B, C, and Δ *functions of the input* x_t, giving the model content-based selectivity – similar in spirit to attention – while keeping the recurrence linear in sequence length. Because this selective recurrence can no longer be computed as a global convolution (the trick S4 used for parallelism), Mamba introduces a hardware-aware parallel scan that recovers GPU-efficient training. Mamba-2 pushes this further by showing that a restricted form of the selective SSM is mathematically equivalent to a form of linear attention ("structured state space duality"), unifying the two families and enabling an even faster matrix-multiplication-based algorithm.

The material is built around three ideas learners can implement themselves: (1) discretizing a continuous linear system, (2) making that system input-selective, and (3) the scan algorithm that parallelizes an ostensibly sequential recurrence.

2 Leveling and prerequisite knowledge

Target level: advanced undergraduate / early graduate. Learners should be comfortable with basic linear algebra (matrix-vector products, eigenvalues), have implemented a standard RNN or attention layer at least once, and have basic PyTorch fluency. No prior exposure to control theory or SSMs is assumed – the discretization step is taught from first principles.

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3 Learning objectives and outcomes

By the end of the material, a learner will be able to:

- **Remember/Understand:** state the recurrence form of a discretized SSM and explain why naive SSMs are linear-time but non-selective.
- **Understand:** explain, in their own words, what "selectivity" adds to an SSM and why it breaks the convolutional-parallelism trick used in S4.
- **Apply:** implement a minimal selective-scan layer in PyTorch and run it on a toy sequence task.
- **Analyze:** compare a selective SSM against self-attention and a vanilla RNN on compute cost, memory, and long-range recall, and interpret the resulting plots.
- **Evaluate/Create:** extend the provided notebook with a variant scan (e.g. bidirectional, as used in vision SSMs) and evaluate whether it helps on a given task.

4 Linked papers

The concept is in active use at the frontier, evidenced by:

1. A. Gu and T. Dao. *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*. COLM, 2024 (arXiv:2312.00752, 2023). Introduces the selective SSM and hardware-aware scan.
2. T. Dao and A. Gu. *Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality*. ICML, 2024. Introduces Mamba-2 and the SSM–attention duality.
3. Y. Liu, Y. Tian, Y. Zhao, et al. *VMamba: Visual State Space Model*. NeurIPS, 2024. Adapts selective scanning to non-sequential (2D image) data via multi-directional scans.
4. F. Rezaei Jafari, G. Montavon, K.-R. Müller, and O. Eberle. *MambaLRP: Explaining Selective State Space Sequence Models*. NeurIPS, 2024. Studies interpretability and faithfulness of explanations for Mamba models.

5 Teaching materials summary

The submission includes:

- **Notebook 1 – From RNN to SSM:** builds a plain linear recurrence, then discretizes a continuous system with a step size Δ, verifying the discretization numerically against a closed-form solution.
- **Notebook 2 – Adding selectivity:** makes B, C, Δ functions of the input, reproducing the core Mamba block, with visualizations of how the hidden state selectively retains or forgets information across a synthetic "selective copying" task.
- **Notebook 3 – Scan vs. sequential loop:** implements a naive sequential scan and a parallel (associative) scan of the same recurrence, benchmarking wall-clock time to build intuition for why the scan algorithm matters.
- **Slides:** a self-contained deck covering the motivation (attention's quadratic cost), the S4-to-Mamba story, the selection mechanism, and a side-by-side comparison table of SSMs vs. attention vs. RNNs.
- **Short exercise set:** five short exercises (e.g. derive the discretization of a 1D system by hand; predict scan output shapes; modify the selective-copying task and predict what breaks) with solutions provided separately.

All code will be provided directly in the submission ZIP (not only linked) and runs on CPU for the toy tasks used in Notebooks 1–2, with an optional GPU path for the benchmark in Notebook 3.